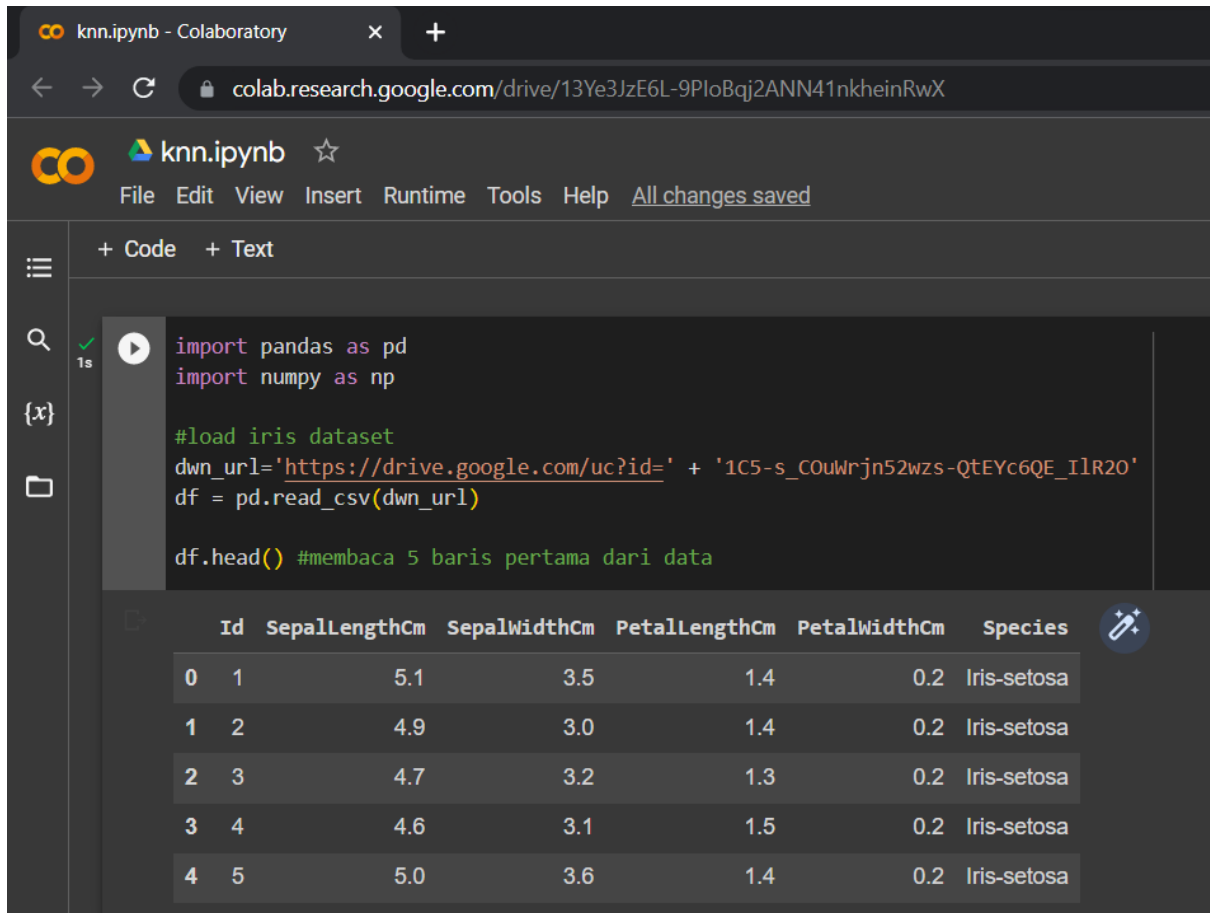


1.



The screenshot shows a Google Colab notebook interface. The browser address bar displays the URL: `colab.research.google.com/drive/13Ye3JzE6L-9PloBqj2ANN41nkheinRwX`. The notebook's title bar indicates it is a `knn.ipynb` file. The menu bar includes options like File, Edit, View, Insert, Runtime, Tools, Help, and a status message "All changes saved".

The code cell contains the following Python code:

```
import pandas as pd
import numpy as np

#load iris dataset
dwn_url='https://drive.google.com/uc?id=' + '1C5-s_COuWrjn52wzs-QtEYc6QE_IIR20'
df = pd.read_csv(dwn_url)

df.head() #membaca 5 baris pertama dari data
```

The output of the code is a table showing the first five rows of the Iris dataset:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

2.

knn.ipynb - Colaboratory

colab.research.google.com/drive/13Ye3JzE6L-9PloBqj2ANN41nkheinRwX#scrollTo=SMW9BkEvJKu-

knn.ipynb

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+ Code + Text

```
[2] # menampilkan unique value pada kolom species
df.Species.unique()
```

```
array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)
```

```
[3] df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype  
---  --
 0   Id              150 non-null   int64  
 1   SepalLengthCm   150 non-null   float64
 2   SepalWidthCm    150 non-null   float64
 3   PetalLengthCm   150 non-null   float64
 4   PetalWidthCm    150 non-null   float64
 5   Species         150 non-null   object  
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

knn.ipynb - Colaboratory

colab.research.google.com/drive/13Ye3JzE6L-9PloBqj2ANN41nkheinRwX#scrollTo=SMW9BkEvJKu-

knn.ipynb

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```
[4] df.describe() #menampilkan summary statistic data
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
[5] # menampilkan jumlah baris dan kolom data
df.shape
```

```
(150, 6)
```

3.

knn.ipynb - Colaboratory

colab.research.google.com/drive/13Ye3JzE6L-9PIoBqj2ANN41nkheinRwX#scrollTo=SMW9BkEvJKu-

knn.ipynb

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```
#menghapus kolom id
df = df.drop('Id', axis = 1)

df
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 5 columns

knn.ipynb - Colaboratory

colab.research.google.com/drive/13Ye3JzE6L-9PloBqj2ANN41nkheinRwX#scrollTo=SMW9BkEvJKu-

knn.ipynb

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```
df = df.sample(150).reset_index(drop = True)
df
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	4.6	3.4	1.4	0.3	Iris-setosa
1	5.7	2.6	3.5	1.0	Iris-versicolor
2	5.7	2.9	4.2	1.3	Iris-versicolor
3	4.6	3.2	1.4	0.2	Iris-setosa
4	6.9	3.1	5.1	2.3	Iris-virginica
...	...	...	...	...	...
145	5.6	3.0	4.1	1.3	Iris-versicolor
146	5.4	3.9	1.3	0.4	Iris-setosa
147	4.9	3.1	1.5	0.1	Iris-setosa
148	5.9	3.0	5.1	1.8	Iris-virginica
149	4.5	2.3	1.3	0.3	Iris-setosa

150 rows x 5 columns

knn.ipynb - Colaboratory

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knn.ipynb

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```
# melakukan fold pada data
fold1 = (df.iloc[0:50].reset_index(drop=True), df.iloc[50:150].reset_index(drop=True))
fold2 = (df.iloc[50:100].reset_index(drop=True), pd.concat([df.iloc[0:50], df.iloc[100:150]]).reset_index(drop=True))
fold3 = (df.iloc[100:150].reset_index(drop=True), df.iloc[0:100].reset_index(drop=True))

test, train = fold2
print(train)
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	4.6	3.4	1.4	0.3	Iris-setosa
1	5.7	2.6	3.5	1.0	Iris-versicolor
2	5.7	2.9	4.2	1.3	Iris-versicolor
3	4.6	3.2	1.4	0.2	Iris-setosa
4	6.9	3.1	5.1	2.3	Iris-virginica
...	...	...	...	...	...
95	5.6	3.0	4.1	1.3	Iris-versicolor
96	5.4	3.9	1.3	0.4	Iris-setosa
97	4.9	3.1	1.5	0.1	Iris-setosa
98	5.9	3.0	5.1	1.8	Iris-virginica
99	4.5	2.3	1.3	0.3	Iris-setosa

[100 rows x 5 columns]

Colaboratory interface showing a Jupyter Notebook with a code cell and a data preview.

Code Cell:

```
def norm(df):  
    df = (df - df.min()) / (df.max()-df.min())  
    return df  
  
X = df.drop('Species',axis=1)  
Y = df.Species  
  
X = norm(X)  
X
```

Data Preview:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	0.083333	0.583333	0.067797	0.083333
1	0.388889	0.250000	0.423729	0.375000
2	0.388889	0.375000	0.542373	0.500000
3	0.083333	0.500000	0.067797	0.041667
4	0.722222	0.458333	0.694915	0.916667
...	...	...	...	...
145	0.361111	0.416667	0.525424	0.500000
146	0.305556	0.791667	0.050847	0.125000
147	0.166667	0.458333	0.084746	0.000000
148	0.444444	0.416667	0.694915	0.708333
149	0.055556	0.125000	0.050847	0.083333

150 rows x 4 columns

4.

knn.ipynb - Colaboratory

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knn.ipynb

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[10]

#menghitung jarak menggunakan euclidean distance

def euclidean(x1, x2):

return np.sqrt(np.sum((x1-x2)**2))

#euclidean distance dr data baris pertama dan kedua

euclidean(X.iloc[0], X.iloc[1])

0.6451608221957991

[11]

#training knn

def KNN(X_train, y_train, X_test, k):

dist = []

#menghitung distance dr data training dan data testing

for row in range(X_train.shape[0]):

dist.append(euclidean(X_train.iloc[row], X_test))

data = X_train.copy()

data['Dist'] = dist

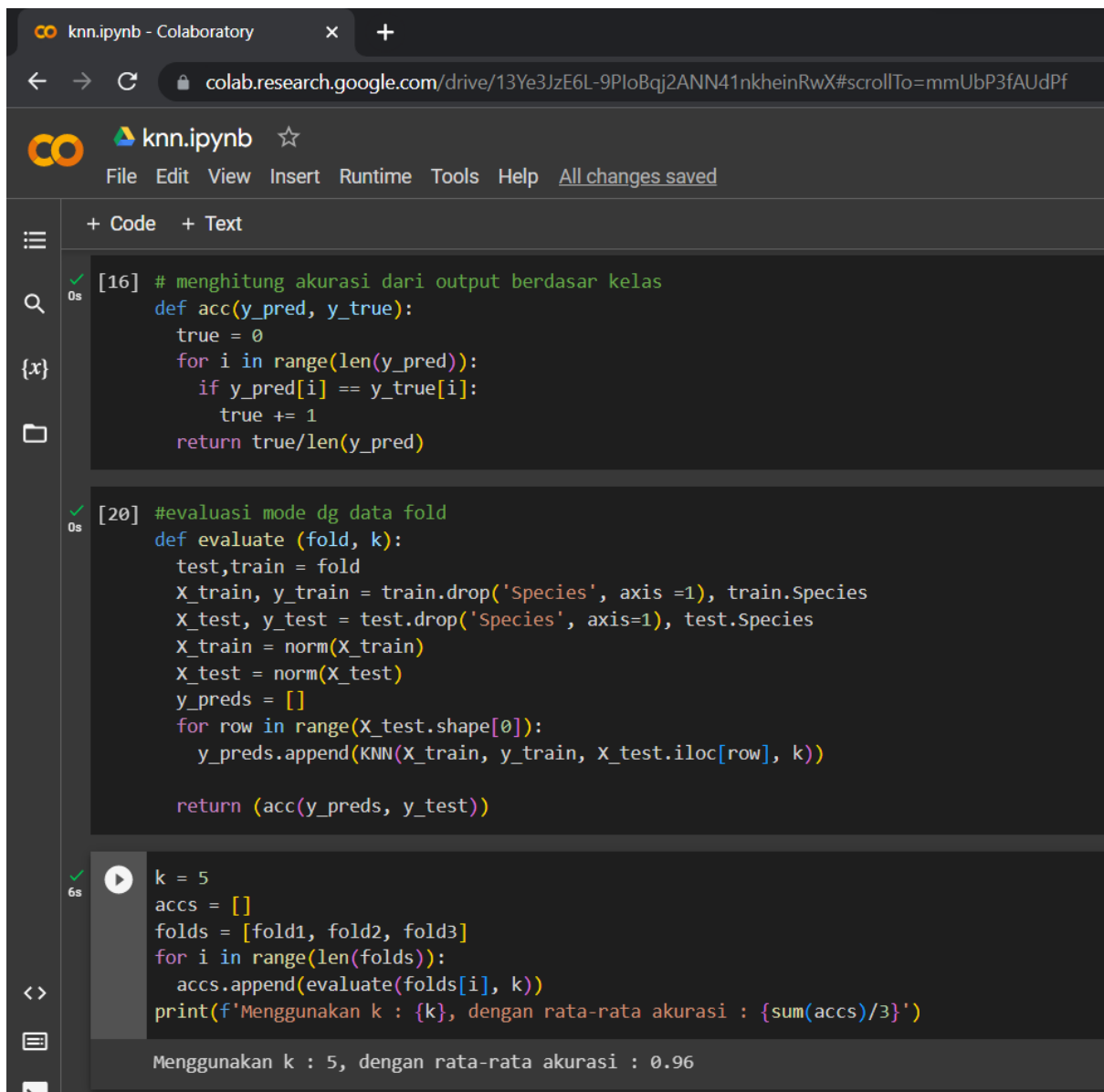
data['Class'] = y_train

data = data.sort_values(by = 'Dist').reset_index(drop=True)

#mengambil label kelas yg sering muncul

y_pred = data.iloc[:k].Class.mode()

return y_pred[0]



```
[16] # menghitung akurasi dari output berdasar kelas
def acc(y_pred, y_true):
    true = 0
    for i in range(len(y_pred)):
        if y_pred[i] == y_true[i]:
            true += 1
    return true/len(y_pred)

[20] #evaluasi mode dg data fold
def evaluate (fold, k):
    test,train = fold
    X_train, y_train = train.drop('Species', axis =1), train.Species
    X_test, y_test = test.drop('Species', axis=1), test.Species
    X_train = norm(X_train)
    X_test = norm(X_test)
    y_preds = []
    for row in range(X_test.shape[0]):
        y_preds.append(KNN(X_train, y_train, X_test.iloc[row], k))

    return (acc(y_preds, y_test))

k = 5
accs = []
folds = [fold1, fold2, fold3]
for i in range(len(folds)):
    accs.append(evaluate(folds[i], k))
print(f'Menggunakan k : {k}, dengan rata-rata akurasi : {sum(accs)/3}')
```

Menggunakan k : 5, dengan rata-rata akurasi : 0.96

LINK GOOGLE COLLAB:

<https://colab.research.google.com/drive/13Ye3JzE6L-9PloBqj2ANN41nkheinRwX?usp=sharing>